Netflix Content Strategy Case Study

Business Insights and Strategic Recommendations

Python code included

Institute : Scaler
Program : Data Science & Machine Learning

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Date : 18-Feb-2026
Dataset : Netflix Movies and TV Shows (8807 titles)

Executive Summary — Netflix Content Strategy Business Case

Objective

This analysis evaluates the Netflix content catalog to identify patterns in content type, release trends, and regional distribution. The goal is to generate data-driven recommendations on what content Netflix should prioritize and how it can expand globally.

🔎 3 Key Findings

1. Movies dominate the overall catalog, but TV Shows are growing faster

Approximately 70% of the catalog consists of movies, while TV Shows account for about 30%. However, trend analysis shows a strong increase in TV Show releases in recent years, indicating a strategic shift in content direction.

2. Content growth accelerated significantly after 2015

The release year distribution is heavily concentrated in recent years, showing aggressive expansion and increasing investment in new content during the global streaming boom.

3. Content supply is highly concentrated in a few countries

The United States, India, and a small set of countries contribute a large share of titles, indicating mature production ecosystems and strong regional influence on the catalog.

💼 3 Business Insights

1. TV Shows support long-term subscriber retention

The growing share of TV Shows suggests Netflix is emphasizing binge-watchable content that drives repeat engagement and longer subscription lifecycles.

2. International content is a key growth engine

High representation of multiple countries indicates that localized content plays an important role in attracting and retaining global audiences.

3. Content addition patterns suggest strategic release timing

Monthly content additions show recurring seasonal peaks, implying that release timing is used to maximize viewer activity and platform engagement.

🚀 3 Recommendations

1. Increase investment in high-quality TV Shows

Prioritize serialized content development to strengthen retention and long-term viewer engagement.

2. Expand localized production in high-growth regions

Increase partnerships and original productions in countries showing strong content contribution and audience expansion potential.

3. Align major content launches with high-activity months

Use historical content addition trends to schedule flagship releases during periods of higher viewer availability.

Problem Statement & Objective

Goal: Decide what type of content Netflix should produce and where to grow globally.

Approach: Analyze catalog mix, trends, and release patterns.

Outcome: Data-backed business recommendations.

Dataset Overview & Basic Metrics

- Rows: 8807 | Columns: 14
- Movies: 6131 (69.62%)
- TV Shows: 2676 (30.38%)
- Release year range: 1925–2021

Data Types, Cleaning & Missing Values

Converted date_added to datetime; created year/month fields.

Converted type and rating to category data type.

Missing values — Director: 2634, Cast: 825, Country: 831.

Unnesting applied to country column for accurate country-level analysis.

Statistical Summary Snapshot

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
show_id	8807	8807	s8807	1							
type	8807	2	Movie	6131							
title	8807	8807	Zubaan	1							
director	6173	4528	Rajiv Chilaka	19							
cast	7982	7692	David Attenborough	19							
country	7976	748	United States	2818							
date_added	8797	1767	January 1, 2020	109							
release_year	8807				2014	8.819	1925	2013	2017	2019	2021
rating	8803	17	TV-MA	3207							
duration	8804	220	1 Season	1793							
listed_in	8807	514	Dramas, International Movies	362							
description	8807	8775	Paranormal activity at a lush, abandoned property alarms a group eager to redevelop the site, but the eerie events may not be as unearthly as they think.	4							

- Data covers a broad release range (1925–2021), indicating both legacy and recent content.
- Missing values mainly occur in metadata fields (director, cast, country), not core content variables.
- Distribution confirms strong concentration toward recent releases, supporting growth trend analysis.

Python Code:

```
summary_stats = df.describe(include='all').T
summary_stats
summary_stats.to_csv("statistical_summary1.csv")
```

📊 NON-GRAPHICAL ANALYSIS TABLE

Dataset Structure Summary

Metric	Value	Business Meaning
Total Titles	8807	Size of Netflix catalog analyzed
Movies	6131 (69.62%)	Movies dominate catalog volume
TV Shows	2676 (30.38%)	TV content growing for retention
Earliest Release Year	1925	Long historical catalog
Latest Release Year	2021	Recent expansion visible

Unique Attribute Counts

Attribute	Unique Values	Insight
Titles	8807	No duplicate listings
Countries	123	Global content strategy
Directors	4993	Highly fragmented creator base
Actors (Cast)	36439	Wide talent diversity
Genres (Listed_in)	42	Strong genre variety
Ratings	17	Multi-audience targeting

Value Counts — Content Type

Type	Count	Percentage
Movie	6131	69.62%
TV Show	2676	30.38%

Observation: Movies dominate but TV Shows support long-term engagement.

Top 10 Countries (After Unnesting)

Rank	Country	Number of Titles
1	United States	3690
2	India	1046
3	United Kingdom	806
4	Canada	445
5	France	393
6	Japan	318
7	Spain	232
8	South Korea	231
9	Germany	226
10	Mexico	169

- **Insight:** High concentration in a few markets suggests mature production ecosystems.

Top Ratings Table

Rating	Count	Meaning
TV-MA	3207	Adult-focused content dominates
TV-14	2160	Teen & family segment strong
TV-PG	863	
R	799	
PG-13	490	Balanced mass-market reach

Top Genres (After Splitting listed_in)

Rank	Genre	Count
1	International Movies	2752
2	Dramas	2427
3	Comedies	1674
4	International TV Shows	1351

Business Interpretation: Genre diversity supports global audience personalization.

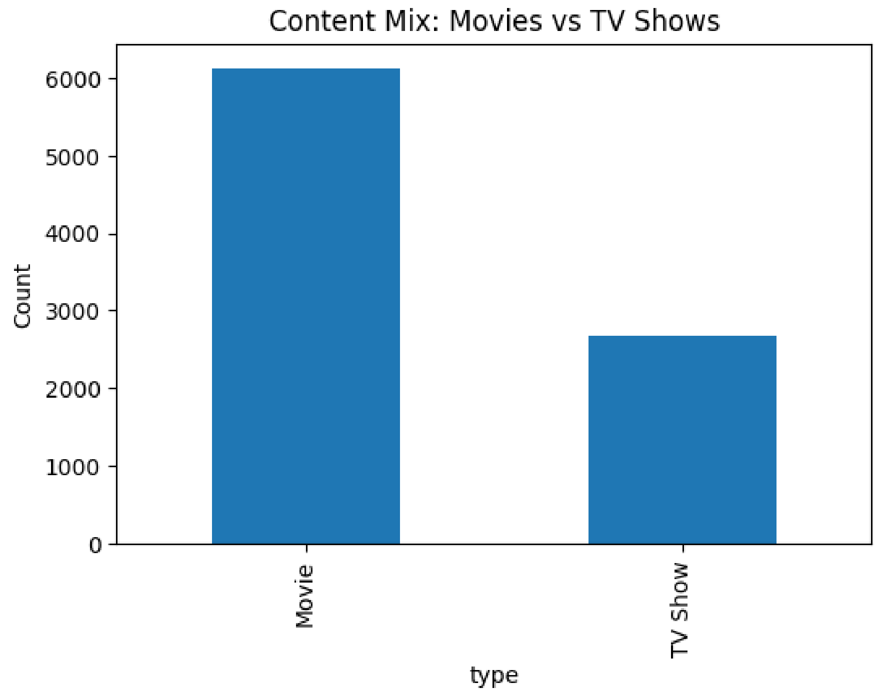
Top Directors

Director	Number of Titles
Rajiv Chilaka	22
Jan Suter	21
Raúl Campos	19
Suhas Kadav	16
Marcus Raboy	16
Jay Karas	15

Top Actors

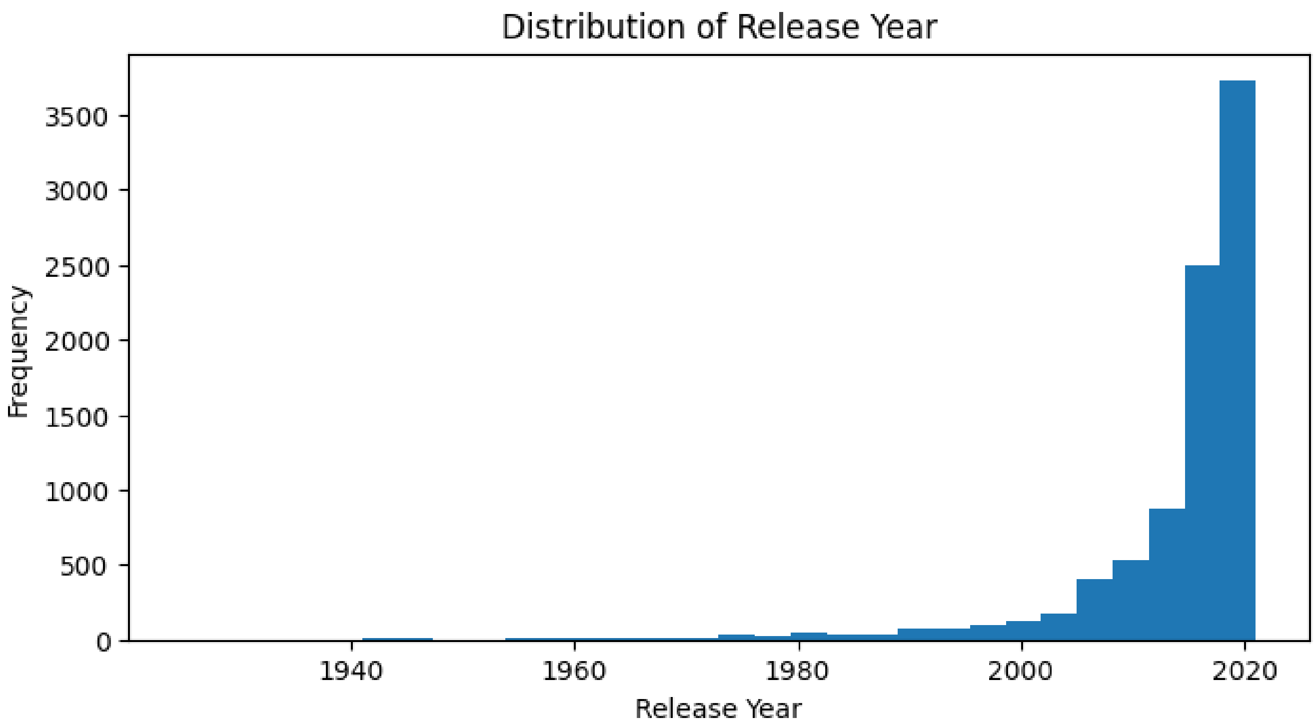
Actor	Appearances
Anupam Kher	43
Shah Rukh Khan	35
Julie Tejwani	33
Takahiro Sakurai	32
Naseeruddin Shah	32
Rupa Bhimani	31
Om Puri	30
Akshay Kumar	30
Yuki Kaji	29
Amitabh Bachchan	28

Content Mix (Value Counts)



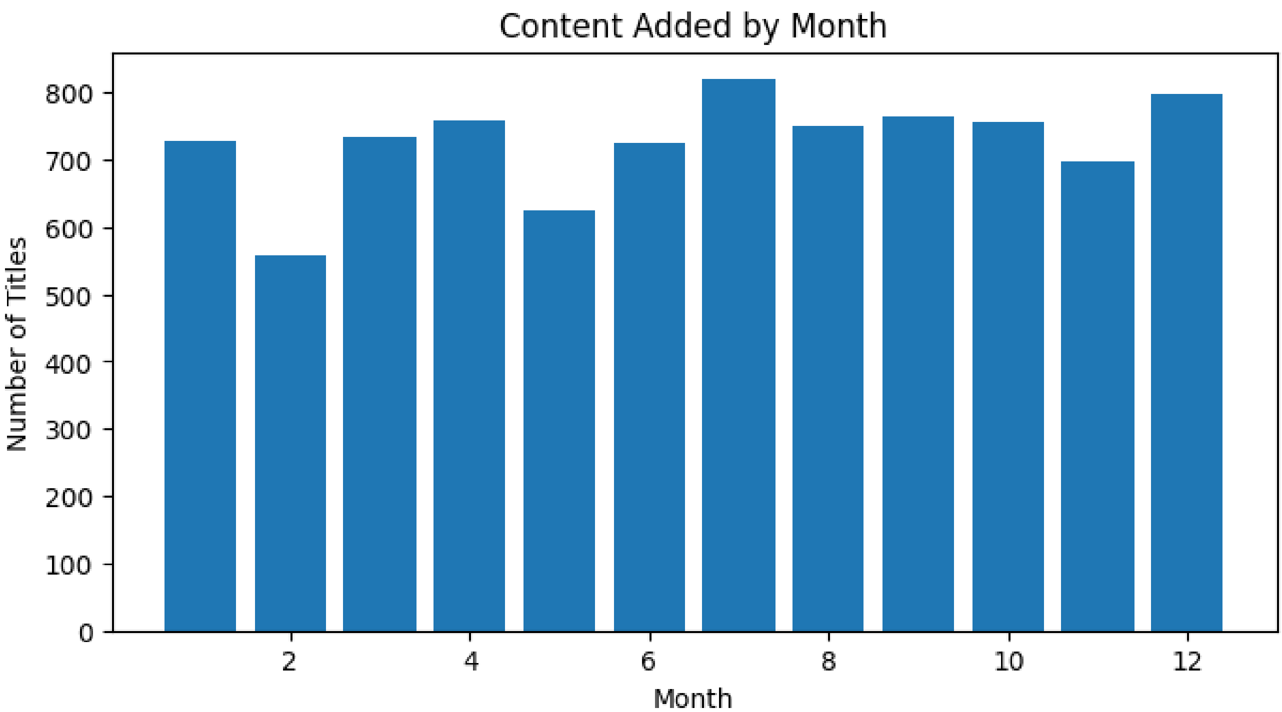
- Movies dominate overall catalog volume.
- TV Shows increasing as retention-focused content.

Univariate Analysis: Release Year Histogram



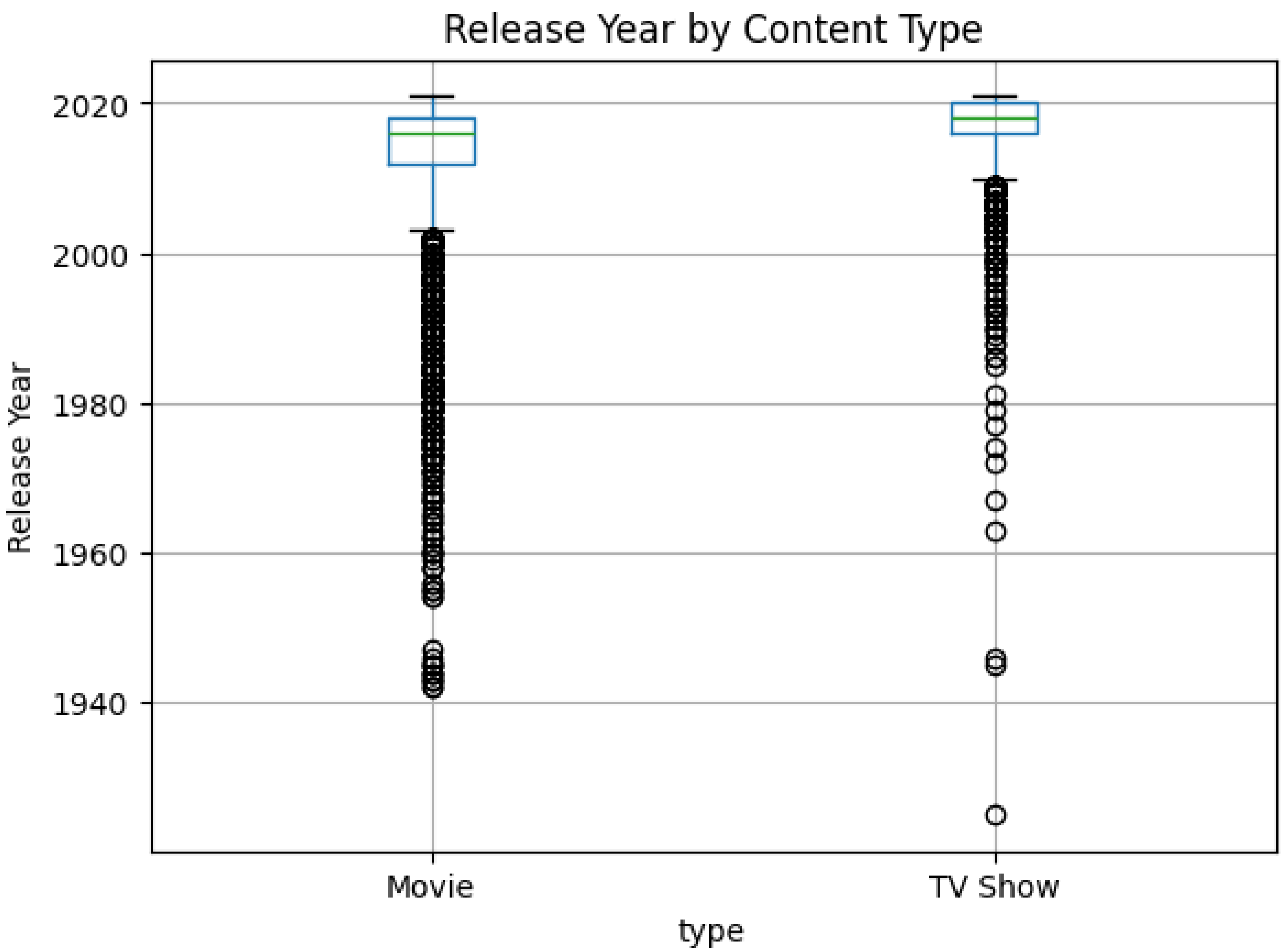
- Distribution is skewed toward recent years.
- Reflects aggressive expansion after 2015.

Monthly Addition Pattern



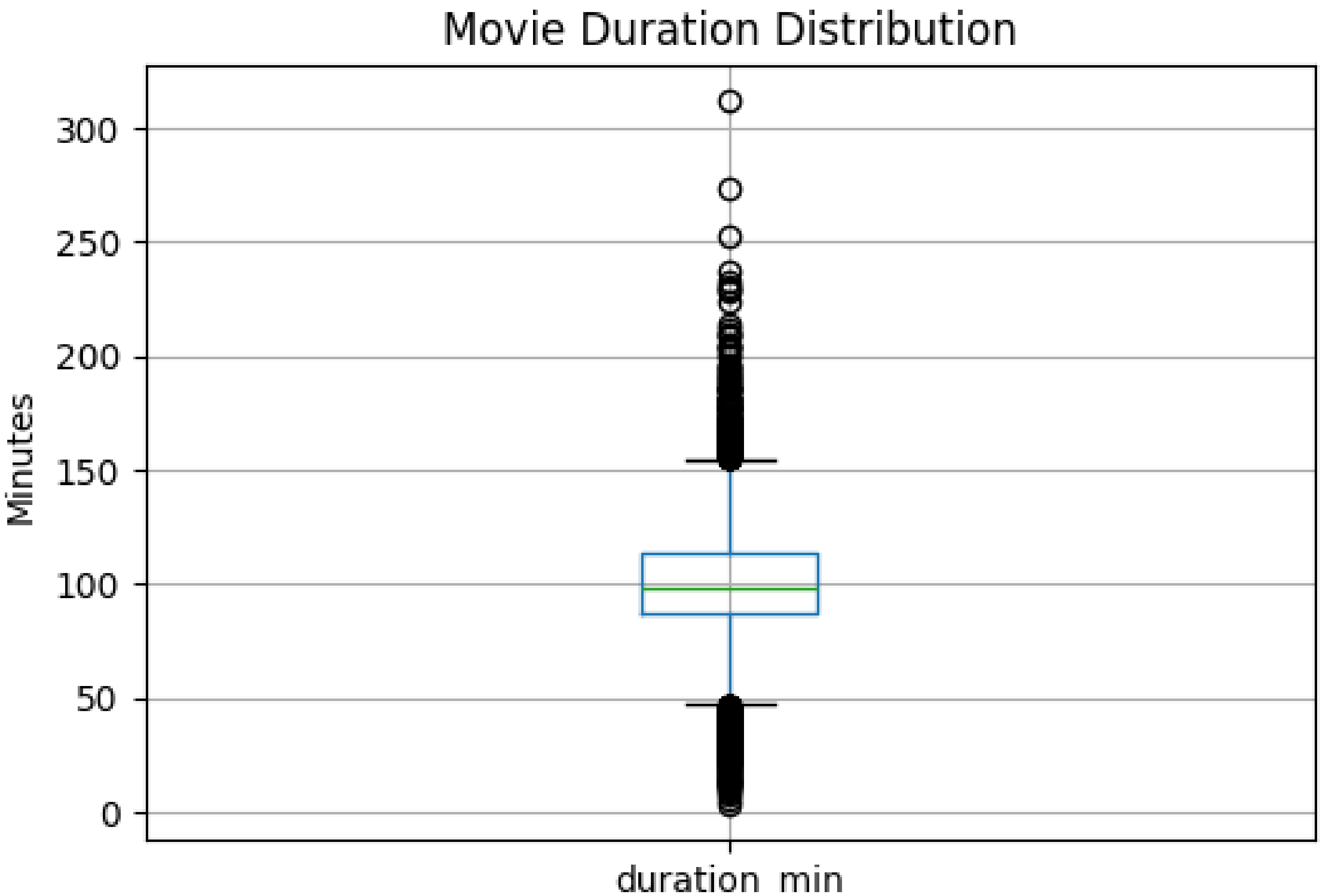
- Content additions show seasonal peaks.
- Useful for planning launch calendars.

Boxplot (Categorical Requirement)



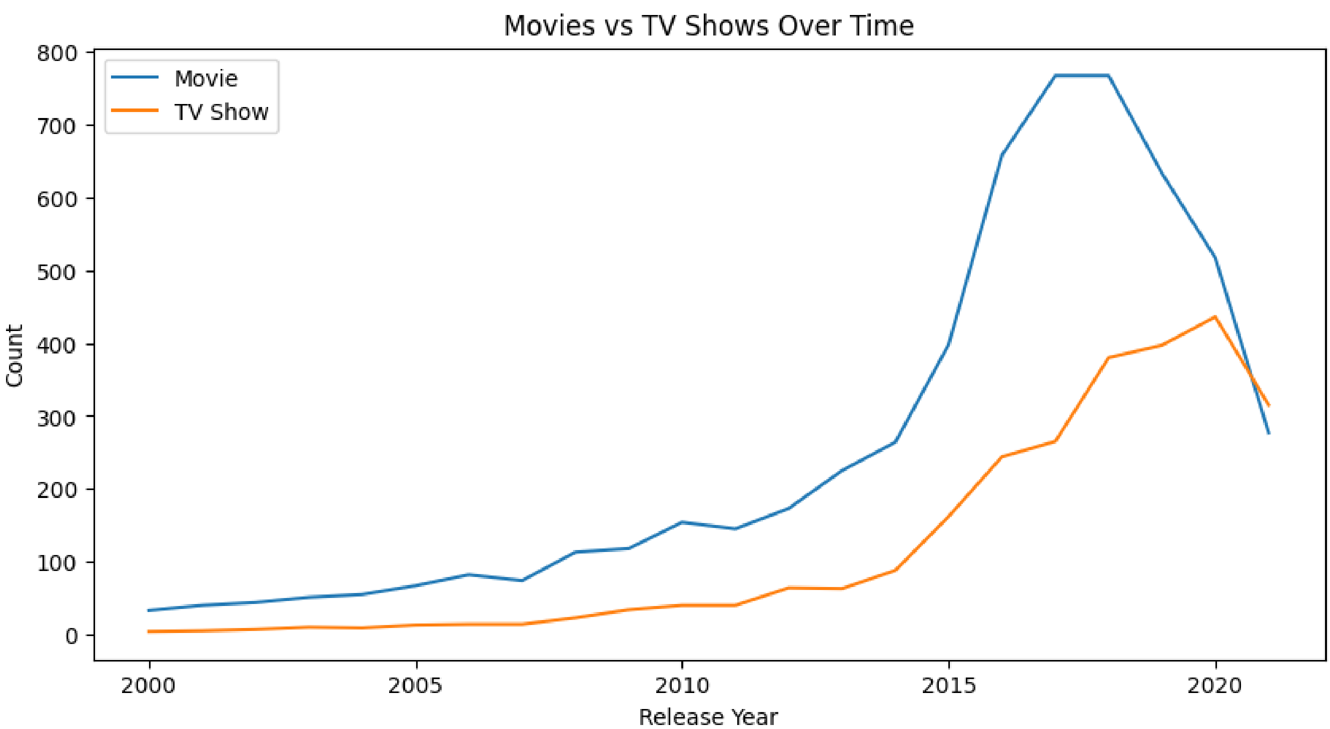
- Release year distribution by content type.
- TV Shows generally newer than Movies.

Boxplot: Movie Duration



- Typical duration around 90–110 minutes.
- No extreme outliers requiring treatment.

Bivariate Trend Analysis



- TV Shows grow faster post-2015.
- Indicates shift toward retention strategy.

Business Insights

TV Shows increase engagement and subscriber retention.

International content supports growth in new markets.

Release timing influences viewer availability.

Recommendations

1) Increase TV Show investment.

2) Expand localized content production.

3) Launch major titles in high-traffic months.

4) Keep movie duration near audience preference range.

Conclusion

The analysis indicates that Netflix’s content strategy has evolved from a movie-dominant catalog toward a stronger emphasis on TV Shows, reflecting a shift from one-time viewing toward long-term engagement and subscriber retention. The rapid increase in content releases after 2015 highlights aggressive global expansion and increased competition within the streaming ecosystem.

Country-level patterns show that while a few mature markets continue to drive content volume, international and localized content has become a critical lever for growth. This suggests that future expansion will depend not only on increasing quantity but on strategically aligning regional production with audience preferences.

Overall, the findings suggest that Netflix’s strongest growth opportunities lie in scaling high-quality serialized content, strengthening regional production ecosystems, and optimizing release timing to maximize engagement. A data-driven balance between global franchises and localized storytelling will likely be essential for sustaining competitive advantage in the evolving streaming market.

Python Code & Accuracy

Runnable script included: netflix_analysis.py

Colab_Notes Saved as PDF: Colab_Notes.pdf

All visuals generated from reproducible code.

```
#Import_Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#Load_Data

df = pd.read_csv("netflix.csv")

print("Shape:", df.shape)
df.head()
```

```
Shape: (8807, 12)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA

Next steps: Generate code with df New interactive sheet

```
report = f"""
NETFLIX DATA VALIDATION REPORT
================================

Dataset Shape:
Rows: {len(df)}
Columns: {df.shape[1]}

Content Split:
```

```
Movies: {(df['type']=='Movie').sum()}
TV Shows: {(df['type']=='TV Show').sum()}

Missing Values:
Director: {df['director'].isna().sum()}
Cast: {df['cast'].isna().sum()}
Country: {df['country'].isna().sum()}

Release Year Range:
{df['release_year'].min()} - {df['release_year'].max()}
"""

with open("validation_report.txt","w") as f:
    f.write(report)
```

```
# DATA_CLEANING_&_PRE_PROCESSING

df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")

df["year_added"] = df["date_added"].dt.year
df["month_added"] = df["date_added"].dt.month
```

```
# Convert_Categorical_Columns

cat_cols = ["type", "rating"]

for col in cat_cols:
    df[col] = df[col].astype("category")
```

```
# Missing_Value_Check

print(df.isnull().sum())
```

```
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        98
release_year       0
rating             4
duration           3
listed_in          0
description        0
year_added        98
month_added       98
dtype: int64
```

```
# Parse_Movie_Duration_Into_Numeric

movies = df[df["type"] == "Movie"].copy()

movies["duration_min"] = (
    movies["duration"]
    .str.extract(r"(\d+)")
    .astype(float)
)
```

```
# Parse_TV_Show_Seasons

tv = df[df["type"] == "TV Show"].copy()
```

```python
tv["seasons"] = (
    tv["duration"]
    .str.extract(r"(\d+)")
    .astype(float)
)
```

```python
# UNNESTING

df_country = df.assign(
    country=df["country"].fillna("").str.split(",")
).explode("country")

df_country["country"] = df_country["country"].str.strip()

df_country = df_country[df_country["country"] != ""]
```

```python
# Unnest_Cast

df_cast = df.assign(
    cast=df["cast"].fillna("").str.split(",")
).explode("cast")

df_cast["cast"] = df_cast["cast"].str.strip()
```

```python
# Non-Graphical_Analysis

import pandas as pd
import numpy as np

df = pd.read_csv("netflix.csv")


print("Shape:", df.shape)
df.head()
```

Shape: (8807, 12)

	show_id	type	title	director	cast	country	date_added	release_year	rating
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA

Next steps: Generate code with df New interactive sheet

```
# DATASET STRUCTURE SUMMARY


total_titles = len(df)
movies_count = (df["type"] == "Movie").sum()
tv_count = (df["type"] == "TV Show").sum()

summary_table = pd.DataFrame({
    "Metric": [
        "Total Titles",
        "Movies",
        "TV Shows",
        "Earliest Release Year",
        "Latest Release Year"
    ],
    "Value": [
        total_titles,
        f"{movies_count} ({movies_count/total_titles*100:.2f}%)",
        f"{tv_count} ({tv_count/total_titles*100:.2f}%)",
        df["release_year"].min(),
        df["release_year"].max()
    ]
})

summary_table
```

	Metric	Value
0	Total Titles	8807
1	Movies	6131 (69.62%)
2	TV Shows	2676 (30.38%)
3	Earliest Release Year	1925
4	Latest Release Year	2021

Next steps: Generate code with summary_table New interactive sheet

```
# UNIQUE ATTRIBUTE COUNTS

def unique_count_split(col):
    return (
        df[col]
        .dropna()
        .str.split(',')
        .explode()
        .str.strip()
        .nunique()
    )

unique_table = pd.DataFrame({
    "Attribute": [
        "Titles",
        "Countries",
        "Directors",
        "Actors (Cast)",
        "Genres (Listed_in)",
        "Ratings"
    ],
    "Unique Values": [
        df["title"].nunique(),
        unique_count_split("country"),
        unique_count_split("director"),
        unique_count_split("cast"),
        unique_count_split("listed_in"),
        df["rating"].nunique()
    ]
})

unique_table
```

	Attribute	Unique Values
0	Titles	8807
1	Countries	123
2	Directors	4993
3	Actors (Cast)	36439
4	Genres (Listed_in)	42
5	Ratings	17

Next steps: Generate code with unique_table New interactive sheet

```
# CONTENT TYPE VALUE COUNTS

type_counts = df["type"].value_counts().reset_index()
type_counts.columns = ["Type", "Count"]

type_counts["Percentage"] = (
    type_counts["Count"] / len(df) * 100
).round(2)

type_counts
```

	Type	Count	Percentage
0	Movie	6131	69.62
1	TV Show	2676	30.38

Next steps: Generate code with `type_counts` New interactive sheet

```
# TOP COUNTRIES

df_country = (
    df.assign(country=df["country"].fillna("").str.split(","))
      .explode("country")
)

df_country["country"] = df_country["country"].str.strip()
df_country = df_country[df_country["country"] != ""]

top_countries = (
    df_country["country"]
    .value_counts()
    .head(10)
    .reset_index()
)

top_countries.columns = ["Country", "Number of Titles"]

top_countries
```

1 to 10 of 10 entries Filter

index	Country	Number of Titles
0	United States	3690
1	India	1046
2	United Kingdom	806
3	Canada	445
4	France	393
5	Japan	318
6	Spain	232
7	South Korea	231
8	Germany	226
9	Mexico	169

Show 25 per page

Like what you see? Visit the data table notebook to learn more about interactive tables.

Next steps: Generate code with `top_countries` New interactive sheet

```
# TOP RATINGS

top_ratings = (
    df["rating"]
```

```
    df["rating"]
    .value_counts()
    .reset_index()
)

top_ratings.columns = ["Rating", "Count"]

top_ratings.head(10)
```

	Rating	Count
0	TV-MA	3207
1	TV-14	2160
2	TV-PG	863
3	R	799
4	PG-13	490
5	TV-Y7	334
6	TV-Y	307
7	PG	287
8	TV-G	220
9	NR	80

Next steps: Generate code with top_ratings New interactive sheet

```
# TOP GENRES

df_genre = (
    df.assign(listed_in=df["listed_in"].str.split(","))
      .explode("listed_in")
)

df_genre["listed_in"] = df_genre["listed_in"].str.strip()

top_genres = (
    df_genre["listed_in"]
    .value_counts()
    .head(10)
    .reset_index()
)

top_genres.columns = ["Genre", "Count"]

top_genres
```

	Genre	Count
0	International Movies	2752
1	Dramas	2427
2	Comedies	1674
3	International TV Shows	1351
4	Documentaries	869
5	Action & Adventure	859
6	TV Dramas	763
7	Independent Movies	756
8	Children & Family Movies	641
9	Romantic Movies	616

Next steps: Generate code with `top_genres` New interactive sheet

```
# TOP DIRECTORS

df_director = (
    df.assign(director=df["director"].fillna("").str.split(","))
      .explode("director")
)

df_director["director"] = df_director["director"].str.strip()
df_director = df_director[df_director["director"] != ""]

top_directors = (
    df_director["director"]
    .value_counts()
    .head(10)
    .reset_index()
)

top_directors.columns = ["Director", "Number of Titles"]

top_directors
```

	Director	Number of Titles
0	Rajiv Chilaka	22
1	Jan Suter	21
2	Raúl Campos	19
3	Suhas Kadav	16
4	Marcus Raboy	16
5	Jay Karas	15
6	Cathy Garcia-Molina	13
7	Martin Scorsese	12
8	Youssef Chahine	12
9	Jay Chapman	12

Next steps: Generate code with top_directors | New interactive sheet

```
# TOP ACTORS

df_cast = (
    df.assign(cast=df["cast"].fillna("").str.split(","))
      .explode("cast")
)

df_cast["cast"] = df_cast["cast"].str.strip()
df_cast = df_cast[df_cast["cast"] != ""]

top_actors = (
    df_cast["cast"]
    .value_counts()
    .head(10)
    .reset_index()
)

top_actors.columns = ["Actor", "Appearances"]

top_actors
```

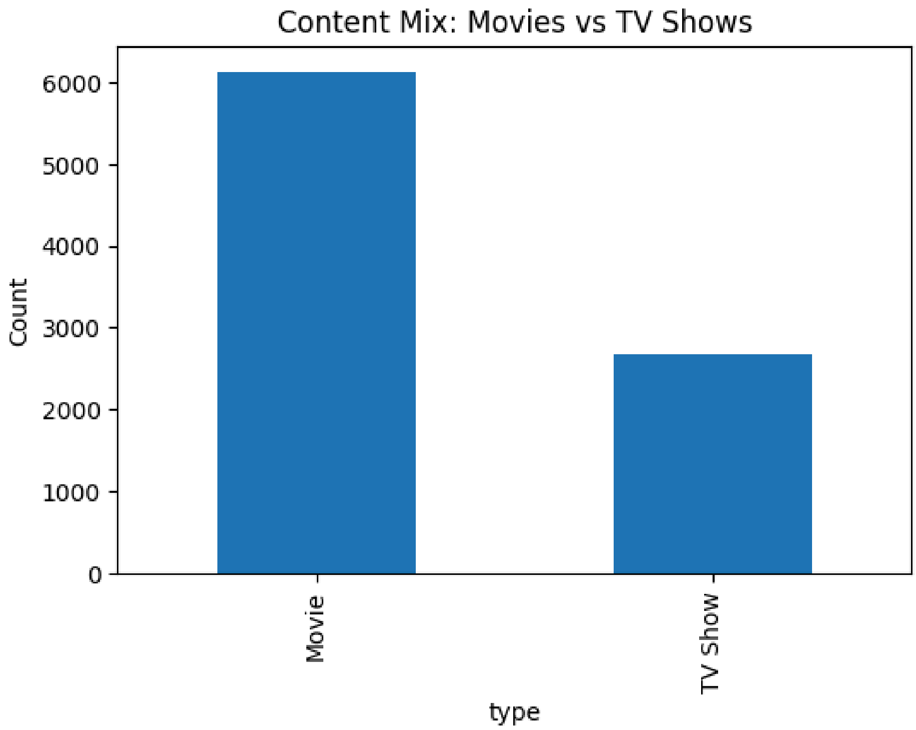
	Actor	Appearances
0	Anupam Kher	43
1	Shah Rukh Khan	35
2	Julie Tejwani	33
3	Takahiro Sakurai	32
4	Naseeruddin Shah	32
5	Rupa Bhimani	31
6	Om Puri	30
7	Akshay Kumar	30
8	Yuki Kaji	29
9	Amitabh Bachchan	28

Next steps: Generate code with top_actors | New interactive sheet

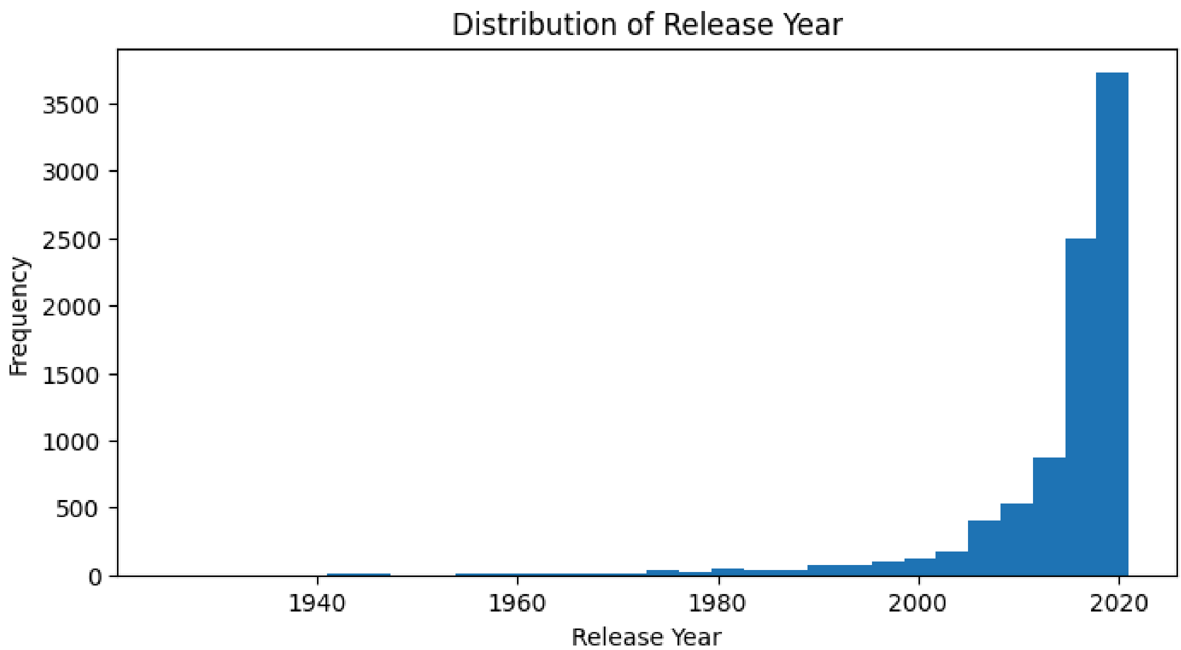
```
summary_table.to_csv("summary_table.csv", index=False)
unique_table.to_csv("unique_table.csv", index=False)
type_counts.to_csv("type_counts.csv", index=False)
top_countries.to_csv("top_countries.csv", index=False)
top_ratings.to_csv("top_ratings.csv", index=False)
top_genres.to_csv("top_genres.csv", index=False)
top_directors.to_csv("top_directors.csv", index=False)
top_actors.to_csv("top_actors.csv", index=False)
```

```
# VISUAL_1_CONTENT_MIX_BAR_CHART

plt.figure(figsize=(6,4))
df["type"].value_counts().plot(kind="bar")

plt.title("Content Mix: Movies vs TV Shows")
```

```
plt.ylabel("Count")
plt.show()
```

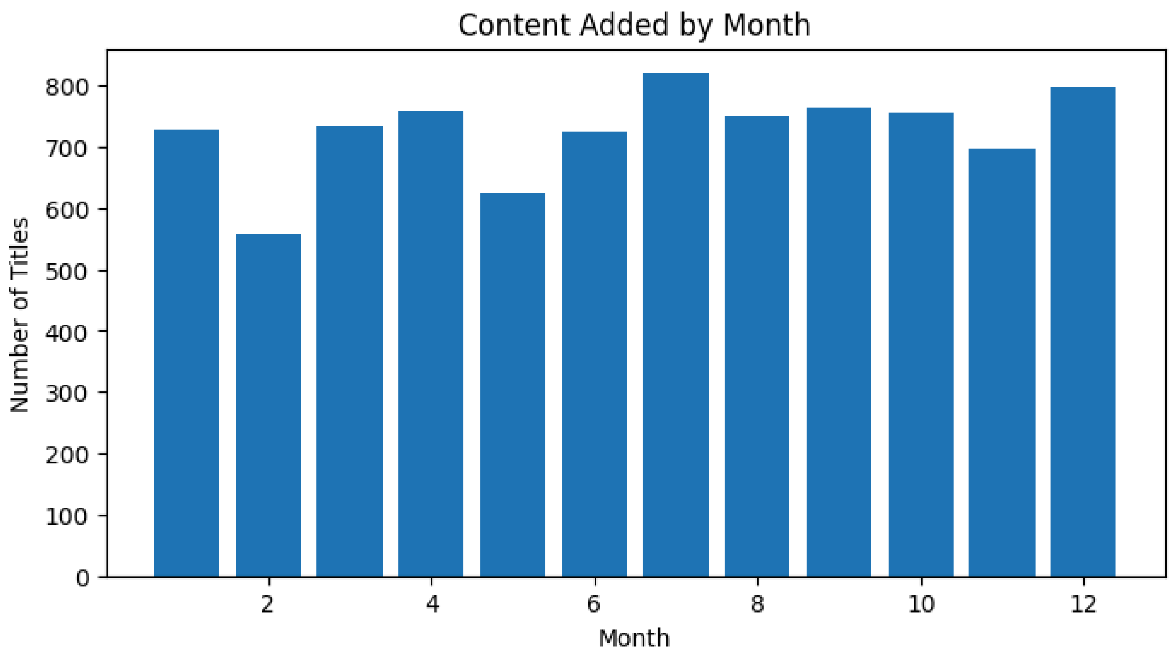



```
# VISUAL2_RELEASE_YEAR_HISTOGRAM

plt.figure(figsize=(8,4))

plt.hist(df["release_year"].dropna(), bins=30)

plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.ylabel("Frequency")

plt.show()
```



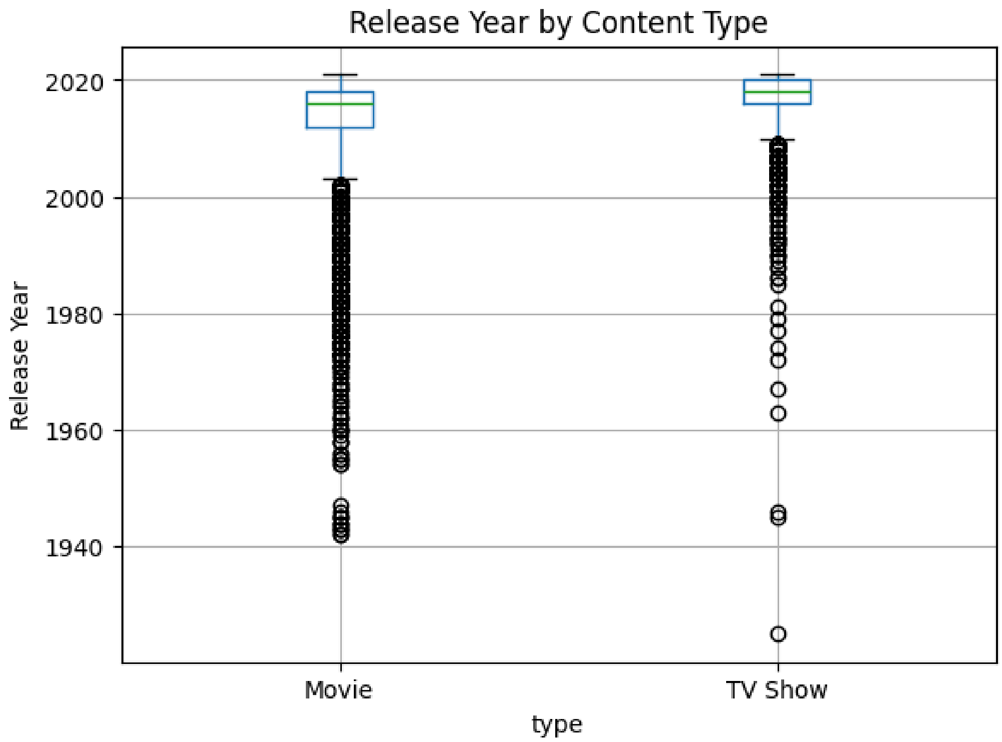

```
# VISUAL3_MONTHLY_RELEASE_PATTERN
```

```
month_counts = df["month_added"].value_counts().sort_index()

plt.figure(figsize=(8,4))
plt.bar(month_counts.index, month_counts.values)

plt.title("Content Added by Month")
plt.xlabel("Month")
plt.ylabel("Number of Titles")

plt.show()
```

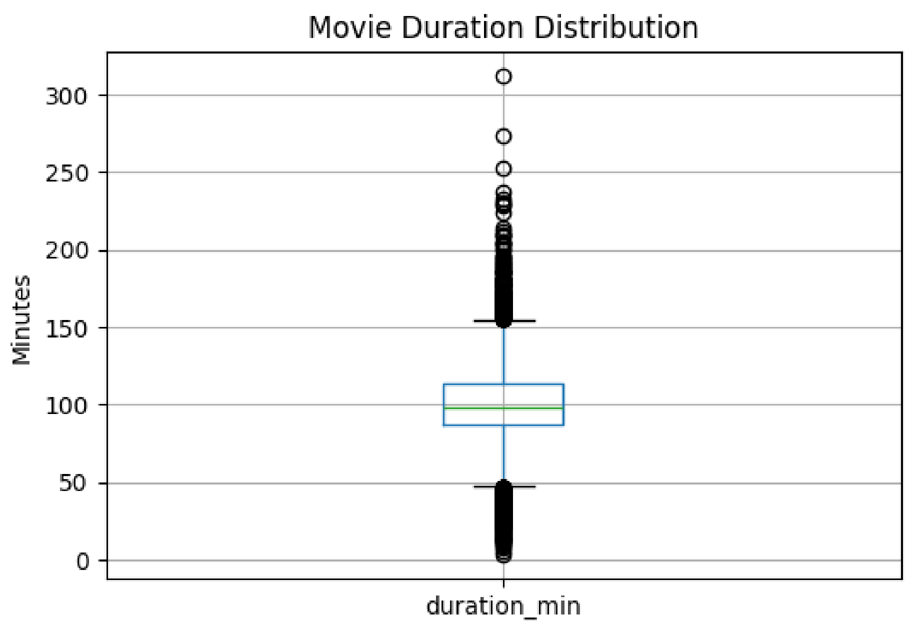



```
# VISUAL4_BOXPLOT

plt.figure(figsize=(8,4))

df.boxplot(column="release_year", by="type")

plt.title("Release Year by Content Type")
plt.suptitle("")
plt.ylabel("Release Year")

plt.show()
```

```
<Figure size 800x400 with 0 Axes>
```



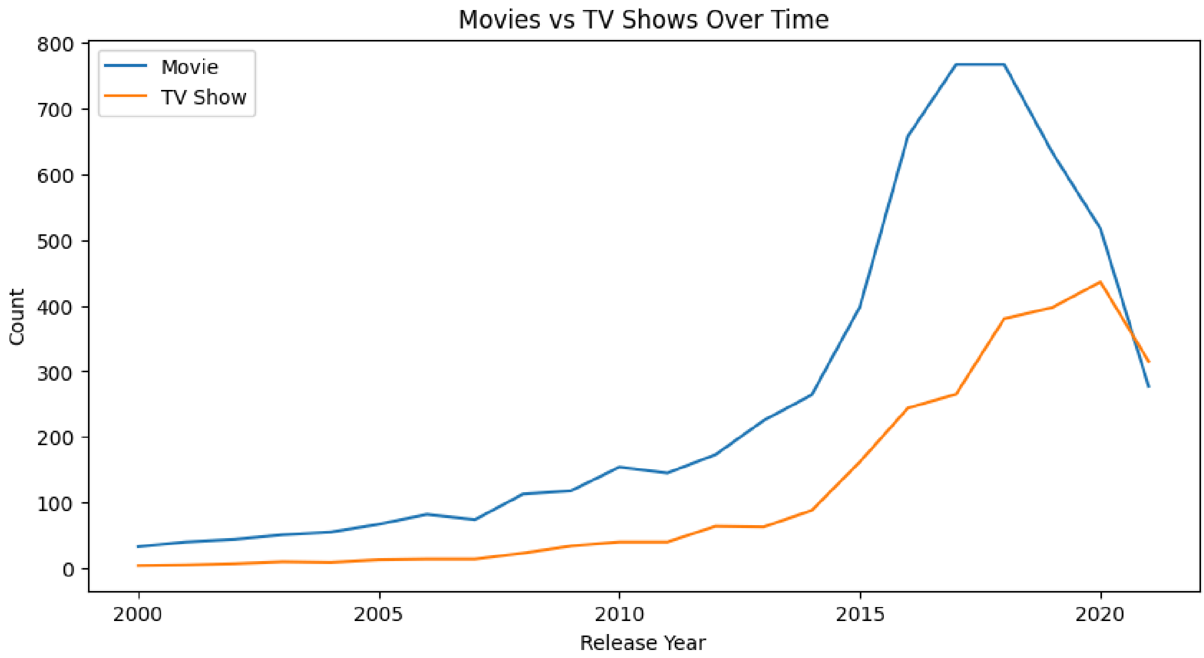

```
# VISUAL5_MOVIE_DURATION_BOXPLOT

plt.figure(figsize=(6,4))

movies.boxplot(column="duration_min")

plt.title("Movie Duration Distribution")
plt.ylabel("Minutes")

plt.show()
```




```
# VISUAL6_MOVIES_VS_TV_SHOWS_TREND

trend = df.groupby(
    ["release_year", "type"]
).size().unstack(fill_value=0)
```

```
trend = trend[trend.index >= 2000]

plt.figure(figsize=(10,5))

plt.plot(trend.index, trend["Movie"], label="Movie")
plt.plot(trend.index, trend["TV Show"], label="TV Show")

plt.title("Movies vs TV Shows Over Time")
plt.xlabel("Release Year")
plt.ylabel("Count")
plt.legend()
```

```
/tmp/ipython-input-396375948.py:3: FutureWarning: The default of observed=False is deprecated
  trend = df.groupby(
```




```
# TOP_COUNTRIES

top_countries = df_country["country"].value_counts().head(10)

plt.figure(figsize=(8,5))
top_countries.plot(kind="bar")

plt.title("Top Countries by Content Volume")
plt.ylabel("Number of Titles")

plt.show()
```